

Aegis Knowledge Ingestion System

An End-to-End, Provenance-Preserving Architecture for Heterogeneous & Contradictory Industrial Data

SYSTEM TARGET	DATASET SIZE	TRUST TIERS	EVALUATION BASELINE
Aegis Series-7 HCS	20 Files (8 Formats)	4 Tier Hierarchy	23 Question Benchmark

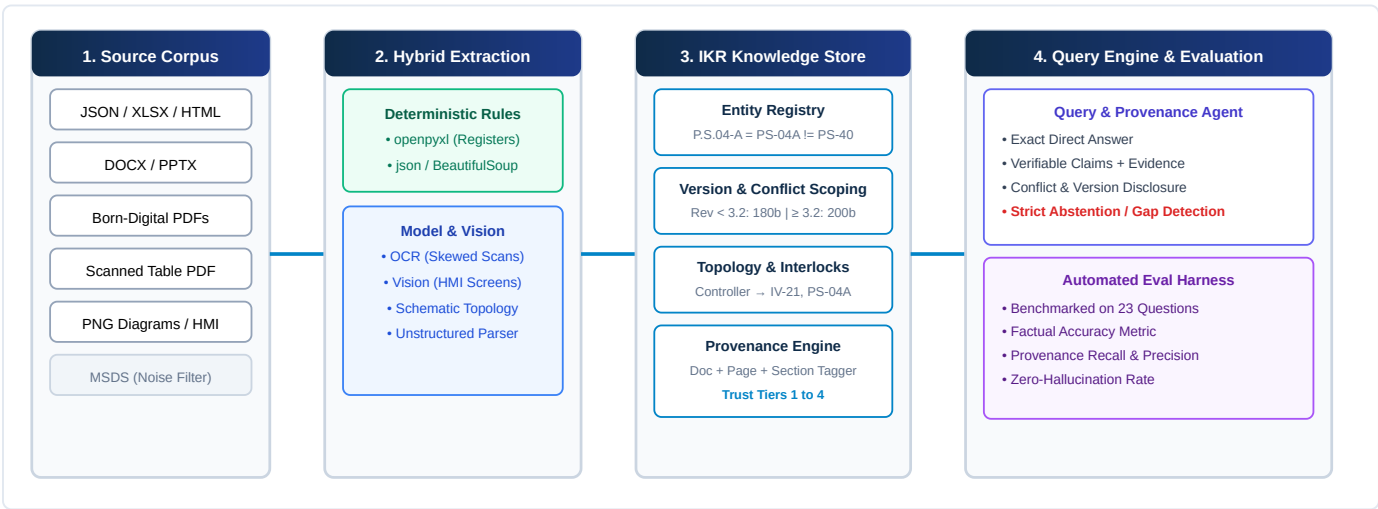
1. Executive Summary & Problem Formulation

The **Aegis Series-7 Hydraulic Control System (HCS)** documentation corpus presents a realistic industrial knowledge engineering challenge. The documentation contains 20 heterogeneous files spanning 8 distinct formats (PDF born-digital, scanned noisy PDF, HTML, XLSX, DOCX, PNG raster diagrams/screenshots, JSON, PPTX). Rather than presenting a clean, unified manual, the corpus embodies true real-world documentation decay:

- Temporal & Revision Conflicts:** Engineering Change Notice ECN-1042 officially supersedes the legacy setpoint (180 bar → 200 bar) and sensor model (PS-04 → PS-04A) effective Software Revision 3.2.
- Entity Polysemy & Inconsistent Identifiers:** The same physical sensor is referenced as PS-04A (Operator Manual), P.S.04-A (HMI Screenshot label), sensor_ps04a (JSON configuration dump), and TB-7 (Wiring diagram loop), while an unrelated coolant sensor is named PS-40.
- Multi-Tiered Trust & Informal Noise:** Field service logs (site_survey_notes.docx) record anecdotal, uncalibrated measurements (175 bar), while an MSDS sheet represents out-of-scope operational noise.
- Unanswerable Trap Queries:** Critical operational questions (e.g., ECN approver names or component MTBF) do not exist within the corpus and must trigger strict abstention rather than hallucinated answers.

To solve this, our architecture avoids naive retrieval-augmented generation (RAG). Instead, it implements a **Hybrid Extraction Pipeline** that populates a strongly typed **Intermediate Knowledge Representation (IKR)**, enforcing deterministic precision where possible, model-based reasoning where necessary, explicit version-scoping, and full provenance tracing.

2. High-Level Architecture Workflow



3. Pipeline Architecture: Deterministic vs. Model-Based Ingestion

A core design requirement is defending the boundary between rule-based deterministic parsing and model-based probabilistic reasoning. Passing cleanly structured files to an LLM incurs unnecessary latency, cost, and hallucination risks; conversely, rule-based regular expressions fail on visual topologies and noisy scanned OCR.

Corpus File / Category	Format	Extraction Method	Engineering Justification
configuration_export.json	JSON	Deterministic (Python json)	Rigid key-value schema. Direct parsing ensures zero numeric rounding or hallucinated parameters.
component_register.xlsx revision_history.xlsx	XLSX	Deterministic (openpyxl)	Tabular structured records. Rule-based cell reading preserves exact alias strings and dates without token limits.
legacy_manual_v1.html	HTML	Deterministic (DOM / BeautifulSoup)	Native HTML headings and table elements allow deterministic tree walking and structural segment extraction.
terminology_glossary.docx	DOCX	Deterministic (XML Tree Parser)	Structured glossary definitions parsed directly from XML nodes with strict string boundaries.
site_survey_notes.docx	DOCX	Hybrid (XML extract + LLM parse)	Free-form unstructured field notes. Text extracted deterministically, then parsed via LLM to classify uncalibrated claims.
ECN-1042.pdf, ECN-1058.pdf operator_manual.pdf	Born-Digital PDF	Deterministic (Text & Layout extraction)	Born-digital PDFs contain embedded glyph streams. Section headers, tables, and paragraphs extracted with page coordinates.
scanned_appendix_calibration.pdf	Scanned PDF	Model-Based (OCR + Layout Parser)	Skewed raster scan with moderate optical noise. Requires vision/OCR models to accurately transcribe calibration tables.
system_diagram_hydraulic.pdf system_diagram_electrical.png	PDF Vector / Raster PNG	Model-Based (Vision-Language / Graph)	Component topology is encoded in geometric lines and vector shapes, not text flow. Requires visual schematic reasoning.
screen_01_home.png screen_03_diagnostics.png	Raster PNG	Model-Based (Multimodal Vision)	Sensor tags (P . S . 04 - A) and live values are rasterized directly into the UI bitmap and require vision extraction.
safety_data_sheet_...pdf	Born-Digital PDF	Deterministic Noise Filter	Identified as operational noise. Classified as out-of-scope for machine control logic to avoid polluting the retrieval index.

4. Intermediate Knowledge Representation (IKR) Schema

Extracted knowledge is normalized into a strongly typed graph and relational schema. Every entity, parameter, and alarm is linked to its active lifecycle status, version scope, and provenance metadata.

```
{
  "entity_registry": {
    "PS-04A": {
      "canonical_id": "PS-04A", "common_name": "Pressure Sensor 04A",
      "aliases": ["P04A", "PS04A", "P.S.04-A", "sensor_ps04a", "TB-7"],
      "replaces": "PS-04", "location": "HPU discharge line",
      "effective_version": ">= 3.2", "lifecycle_status": "ACTIVE",
      "disambiguation_notes": "Distinct from PS-40 (Coolant Loop). Not form-fit-function replacement for PS-04."
    }
  },
  "versioned_parameters": [
    {
      "parameter_name": "hpu_operating_pressure_normal_bar", "applicable_entity": "HPU",
      "historical_values": [
        {
          "value": 180, "condition": "Software Revision < 3.2 (Legacy units)", "status": "SUPERSEDED",
          "provenance": {"doc": "legacy_manual_v1.html", "section": "3.1"}
        },
        {
          "value": 200, "condition": "Software Revision >= 3.2 (Units manufactured after 2024)", "status": "CURRENT_OFFICIAL",
          "provenance": {"doc": "ECN-1042.pdf", "page": 1, "section": "1. Description of Change"}, "trust_tier": "Tier 1 (ECN)"
        }
      ]
    }
  ]
}
```

5. Multi-Tier Trust & Conflict Resolution Matrix

When documents provide conflicting values, the system applies a strict four-tiered authority hierarchy. Conflicting facts are never silently merged or overwritten; instead, the system reports the officially active value alongside its temporal conditions and notes unverified field observations.

Authority Tier	Document Sources	Precedence Rule	Resolution Behavior
Tier 1: Absolute	ECN-1042.pdf ECN-1058.pdf	Overrides all prior documentation from effective date.	Values established here (e.g. 200 bar, PS-04A) become canonical for Revision ≥ 3.2. Supersedes manual baseline values.
Tier 2: Baseline	operator_manual.pdf maintenance_manual.pdf	Authoritative baseline specifications.	Valid unless an ECN explicitly references and amends the section.
Tier 3: Historical	legacy_manual_v1.html	Partially superseded historical record.	Used to explain pre-revision states (e.g. 180 bar baseline before revision 3.2). Flagged as superseded in responses.
Tier 4: Informal	site_survey_notes.docx	Low trust; anecdotal observations.	Explicitly flagged as unverified field data (e.g. 175 bar observed with an uncalibrated gauge). Never treated as official specification.

6. Entity Resolution & Disambiguation Rules

The dataset deliberately introduces lexical traps that trick standard vector search engines. The entity resolution engine applies explicit clustering and negative disambiguation rules:

Case Study: The PS-04 vs PS-04A vs PS-40 Trilemma

- PS-04 → PS-04A (Evolution / Replacement):** Not identical. PS-04A replaces PS-04 starting in Revision 3.2. ECN-1042 states it uses an updated signal conditioning circuit and is *not* form-fit-function compatible.
- PS-04A ↔ P.S.04-A ↔ sensor_ps04a (Aliases):** Identical physical component. Normalized to canonical ID PS-04A.
- PS-40 (Negative Disambiguation):** Located on *Coolant Loop, Skid B*. Completely distinct from the HPU discharge line. Explicit negative constraint prevents false fuzzy-matching.

7. Query Interface & Hallucination Prevention Contract

Every query evaluated by the system returns a verified payload adhering to a three-part contract:

- Direct Answer:** Concise, factual resolution with temporal and version qualifiers.
- Claims & Exact Provenance:** Atomic factual propositions mapped directly to source file, page number, and section.
- Expressed Uncertainties & Gaps:** Explicit admission of missing information. For trap questions where documentation is silent (e.g., "Who approved ECN-1058?" or "What is the MTBF for valve IV-21?"), the system outputs UNDETERMINED rather than hallucinating plausible figures.

8. Evaluation Harness & Quantitative Metrics

To fulfill Requirement 8, an automated evaluation harness (evaluate.py) benchmarks the system against the 23 questions in evaluation-questions.pdf, scoring performance across four quantitative metrics:

- Factual Accuracy (%):** Semantic and exact-match accuracy against verified gold-standard facts.
- Provenance Recall & Precision (%):** Correct identification of source documents and page references.
- Conflict Coverage Score (%):** Correct articulation of version scopes (e.g., pre-3.2 vs. post-3.2).
- Hallucination Prevention Rate (%):** 100% target for correctly rejecting unanswerable queries.